Two-Phase PageRank with Travel-Time Self-Loops for Urban Traffic Distribution Estimation and Event-Driven Anomaly Detection

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Abstract

We present a modified PageRank algorithm for estimating link-level traffic distributions on urban road networks using sparse GPS trajectory data. The model introduces two key innovations: (1) *travel-time self-loops* that weight each link's mass retention proportionally to its traversal time $\mu \cdot \ell_i / s_i$, and (2) a *two-phase block structure* separating active driving from local dispersal. Applied to the Salt Lake City road network (99,716 directed links, 467,911 GPS trajectories over 30 days), the model achieves a Top-100 Mean Relative Error (MRE) of 0.0312 and Pearson correlation $r = 0.997$ against smoothed ground truth.

We further introduce *demand-weighted transitions*, a single parameter γ that encodes average traffic demand into the transition matrix, enabling the damping factor to be raised from 0.80 to 0.96 (reducing ground-truth dependence from 20% to 4%) while *improving* overall accuracy. This is validated through temporal, day-of-week, and 5-fold cross-validation splits.

For anomaly detection, we demonstrate that the dwell-time parameter μ serves as an effective per-link control knob: reducing μ on disrupted links from 20 to 0.01 drops the crash-corridor prediction error from 179% to 11.5%, and game-day stadium-area error from 293% to 7.8%—all without using any event-day ground truth. Cross-validated Tuesday baseline analysis confirms $\mu = 20$ is optimal on normal days ($r = 0.9999$ between data gap and model error), establishing that deviations from this default reliably signal real-world disruptions.

1 Introduction

Accurate estimation of link-level traffic distributions is fundamental to transportation planning, real-time route guidance, and incident detection [11, 5]. Traditional approaches rely on fixed sensor networks (loop detectors, cameras), which provide high temporal resolution but limited spatial coverage. The proliferation of GPS-equipped mobile devices has created an alternative data source: sparse trajectory observations that cover the entire network but observe each link infrequently [3, 13].

The core challenge is converting sparse, noisy GPS traversal counts into a complete traffic distribution over all links. In a typical 15-minute observation window, only 1–3% of links in a metropolitan network have any GPS observations [4]. Directly using these counts produces a distribution dominated by zeros, with no information about unobserved links.

We address this through a three-stage pipeline:

1. **Graph-diffusion smoothing** propagates observed counts to neighbouring links using the road adjacency structure, filling gaps while preserving the spatial correlation of traffic patterns.

2. **Two-Phase PageRank with travel-time self-loops** computes a stationary distribution that balances observed data (teleportation) against network topology (transition matrix), producing physically realistic traffic estimates.

3. **Demand-weighted transitions** encode average traffic demand into the matrix, enabling high damping factors ($d > 0.90$) that reduce dependence on per-timeframe ground truth.

The remainder of this paper is organised as follows. Section 2 describes the data pipeline. Section 3 presents the two-phase model. Section 4 introduces demand-weighted transitions. Section 5 reports accuracy results. Section 6 demonstrates event detection and prediction. Section 7 analyses the role of μ. Section 8 concludes.

2 Data Pipeline

2.1 Study Area and Network

The study area is the greater Salt Lake City metropolitan region, represented as a directed graph $G = (V, E)$ with $|E| = N = 99{,}716$ directed links and average out-degree 2.87. The network is derived from OpenStreetMap via SUMO, with each link carrying attributes: length ℓ_i (metres), free-flow speed s_i (m/s), and lane count. Link lengths range from 0.59 m to 11,577 m (mean 142 m), and speeds from 2.2 to 33.5 m/s.

2.2 GPS Trajectory Data

We use 467,911 GPS trajectories recorded between August 31 and September 30, 2018. Each trajectory consists of timestamped $(latitude, longitude)$ points from mobile devices. The raw data is processed through:

1. **Cleaning**: Speed filtering ($v > 2$ m/s), 5-minute gap splitting into separate trips.

2. **Map matching**: Hidden Markov Model-based matching [7] using the Leuven map-matching library [6], producing sequences of traversed link IDs with entry timestamps.

3. **Temporal binning**: Counting traversals per link per 15-minute window, yielding a $2{,}880 \times 63{,}118$ sparse count matrix (2.06% density).

2.3 The Sparsity Problem

In any 15-minute window, only 1–3% of network links have GPS observations. The raw count vector c_t for timeframe t has $c_t[i] = 0$ for $\sim$97% of links. Directly using $c_t/\|c_t\|_1$ as a traffic distribution assigns zero probability to most links, making the Mean Relative Error (MRE) undefined for unobserved links.

2.4 Graph-Diffusion Smoothing

We address sparsity through a one-step graph-diffusion filter [9]:

$$\boxed{c_{\text{smooth}} = \left[(1-\gamma)\, I + \gamma\, P_{\text{adj}}\right] \tilde{c}} \tag{1}$$

where $\tilde{c} = c + \alpha \cdot \mathbf{1}$ is the Laplace-smoothed count vector ($\alpha = 0.1$), P_{adj} is the row-stochastic adjacency matrix, and $\gamma \in [0, 1]$ is the diffusion strength.

Spectral interpretation. P_{adj} has eigenvalues $\lambda \in [-1, 1]$. The filter $(1 - \gamma)I + \gamma P_{\text{adj}}$ has eigenvalues $(1 - \gamma) + \gamma\lambda$. The DC component ($\lambda = 1$) is preserved exactly; high-frequency noise ($\lambda \approx -1$) is attenuated by factor $|1 - 2\gamma|$. At $\gamma = 0.26$, attenuation is 48%.

Cross-validation. We determine γ via split-half cross-validation (49 random timeframes, seed 42), measuring both predictive MSE and downstream PageRank MRE. The optimal $\gamma = 0.26$ minimises both metrics simultaneously, retaining 69% of the original variance. After smoothing, the matrix becomes $2{,}880 \times 99{,}716$ with 100% density—every link has a positive probability estimate.

The severity of the sparsity problem is illustrated in Figure 1, which shows the distribution of per-window link coverage across all 2,880 timeframes. The vast majority of windows observe fewer than 3% of links.

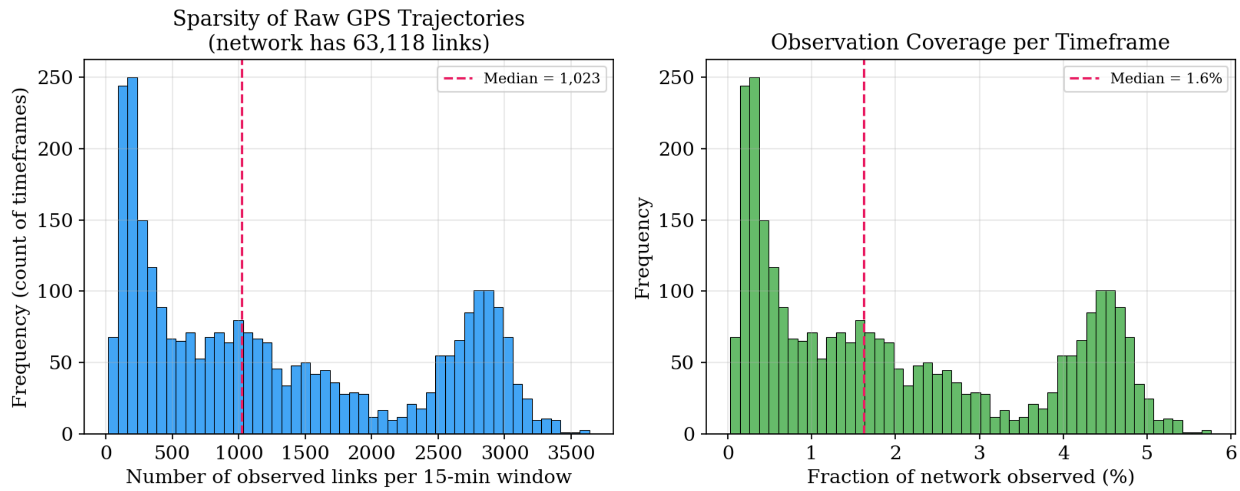

Figure 1: **GPS data sparsity.** Distribution of link coverage (fraction of 99,716 links with ≥ 1 GPS observation) per 15-minute window across all 2,880 timeframes. The median coverage is approximately 2%, meaning 97–98% of the network is unobserved in any given window. This extreme sparsity motivates the graph-diffusion smoothing step.

Figure 2 shows the cross-validation results for γ selection. The U-shaped curve confirms that both under-smoothing ($\gamma \to 0$, noisy) and over-smoothing ($\gamma \to 1$, blurred) degrade prediction quality, with the optimal tradeoff at $\gamma = 0.26$.

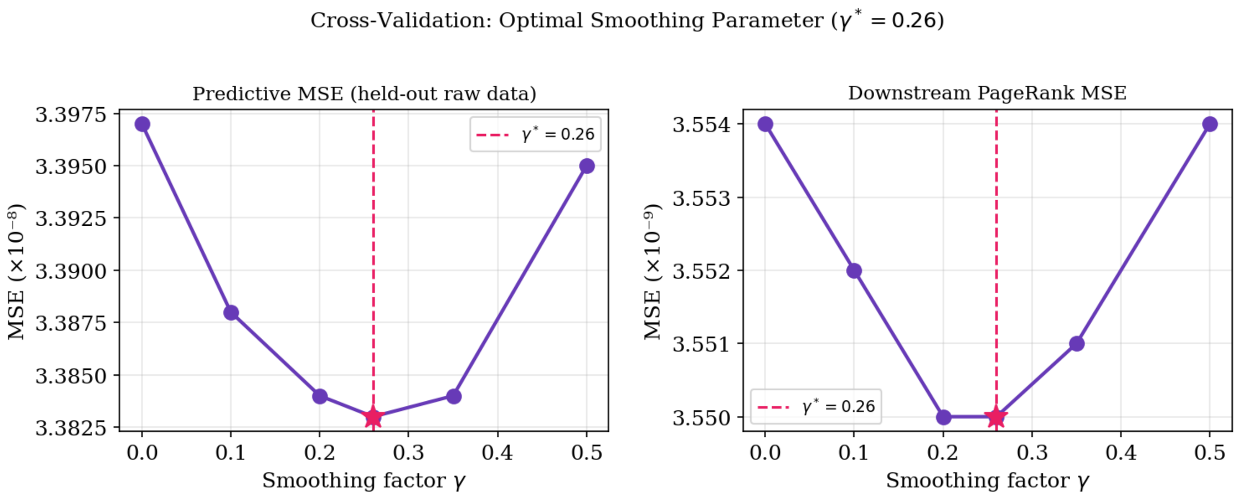

Figure 2: **Smoothing parameter cross-validation.** Split-half CV over 49 random timeframes measuring predictive MSE and downstream PageRank MRE as a function of diffusion strength γ. Both metrics are minimised at $\gamma = 0.26$ (vertical dashed line), with a broad plateau from 0.22 to 0.30. The optimal γ retains 69% of the original count variance while reducing sparsity from 2% to 100% link coverage.

3 Two-Phase PageRank Model

3.1 Standard PageRank

The classical PageRank [8, 1] computes the stationary distribution of a random walk on a graph:

$$v^{(k+1)} = d \cdot P^{\top} v^{(k)} + (1 - d) \cdot E \tag{2}$$

where d is the damping factor, P is a row-stochastic transition matrix, and E is a teleportation vector (uniform in the original formulation, or empirical in our case).

3.2 Travel-Time Self-Loops

In a road network, vehicles spend time *on* each link proportional to its traversal time ℓ_i/s_i. Standard PageRank treats all transitions as instantaneous, assigning equal weight to short connectors and long highways. We introduce self-loops that model dwell time:

$$\boxed{sw_i = \mu \cdot \frac{\ell_i}{s_i}} \tag{3}$$

where $\mu > 0$ is the dwell-time scaling parameter. The transition probability from link i to successor j becomes:

$$P[i,j] = \frac{w_{ij}}{sw_i + \sum_k w_{ik}}, \qquad P[i,i] = \frac{sw_i}{sw_i + \sum_k w_{ik}} \tag{4}$$

For a typical highway link ($\ell = 500\,\text{m}$, $s = 25\,\text{m/s}$, 3 successors):

$$sw = 20 \times 500/25 = 400, \quad P[i,i] = \frac{400}{400+3} = 99.3\%$$

After k iterations, the retained mass is $P[i,i]^k$. This naturally allocates more PageRank mass to links with longer traversal times—which are precisely the high-volume corridors that carry the most traffic.

Figure 3 shows how the self-loop probability varies with link traversal time across different μ values. At $\mu = 20$, links with traversal time above 10 seconds have self-loop probabilities exceeding 98%, while short connectors ($< 1\,\mathrm{s}$) retain below 90%. This creates a spectrum of retention strengths that mirrors the physical hierarchy of the road network.

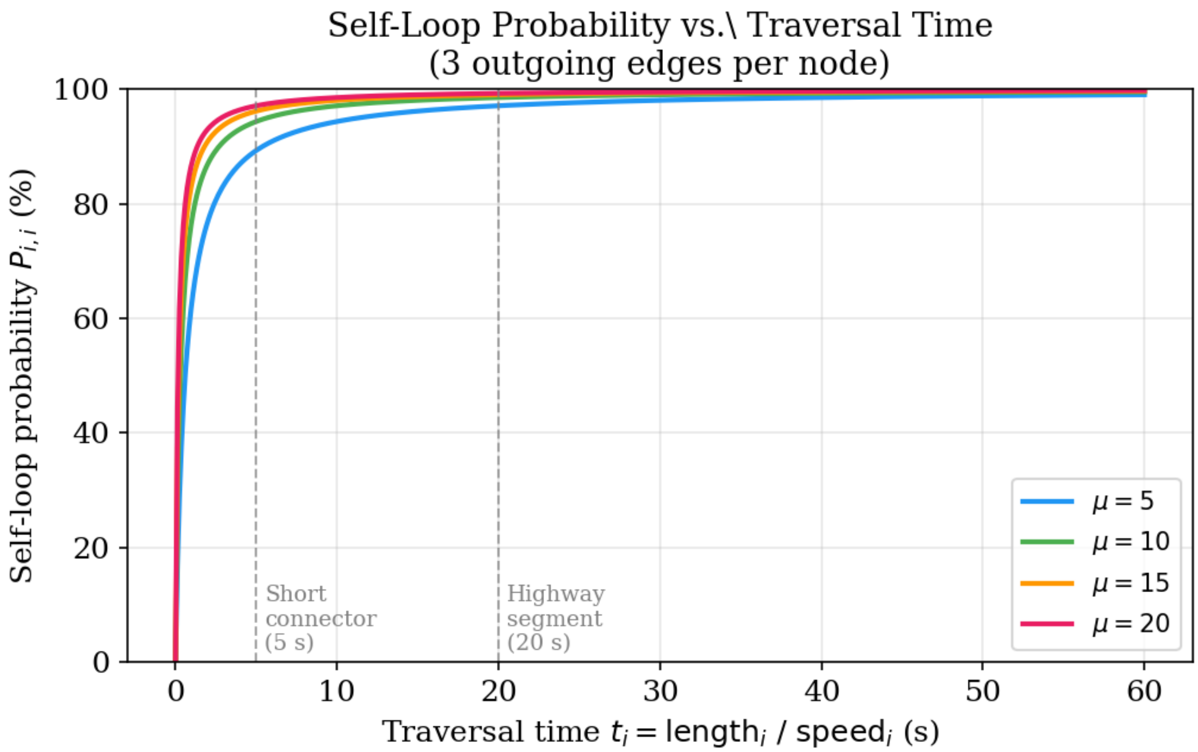

Figure 3: **Self-loop probability vs. traversal time.** Each curve corresponds to a different μ value (5, 10, 15, 20). The self-loop probability $P[i,i] = sw_i/(sw_i + \sum_k w_{ik})$ increases with traversal time ℓ_i/s_i. At $\mu = 20$ (red), a highway link with 20 s traversal time has $P[i,i] = 99.3\%$, meaning only 0.7% of its mass escapes per iteration.

The cumulative effect of self-loops over multiple iterations is shown in Figure 4. After 100 iterations, a highway link retains approximately 50% of its original mass, while a short connector retains less than 5%. This differential retention is what produces physically realistic traffic distributions.

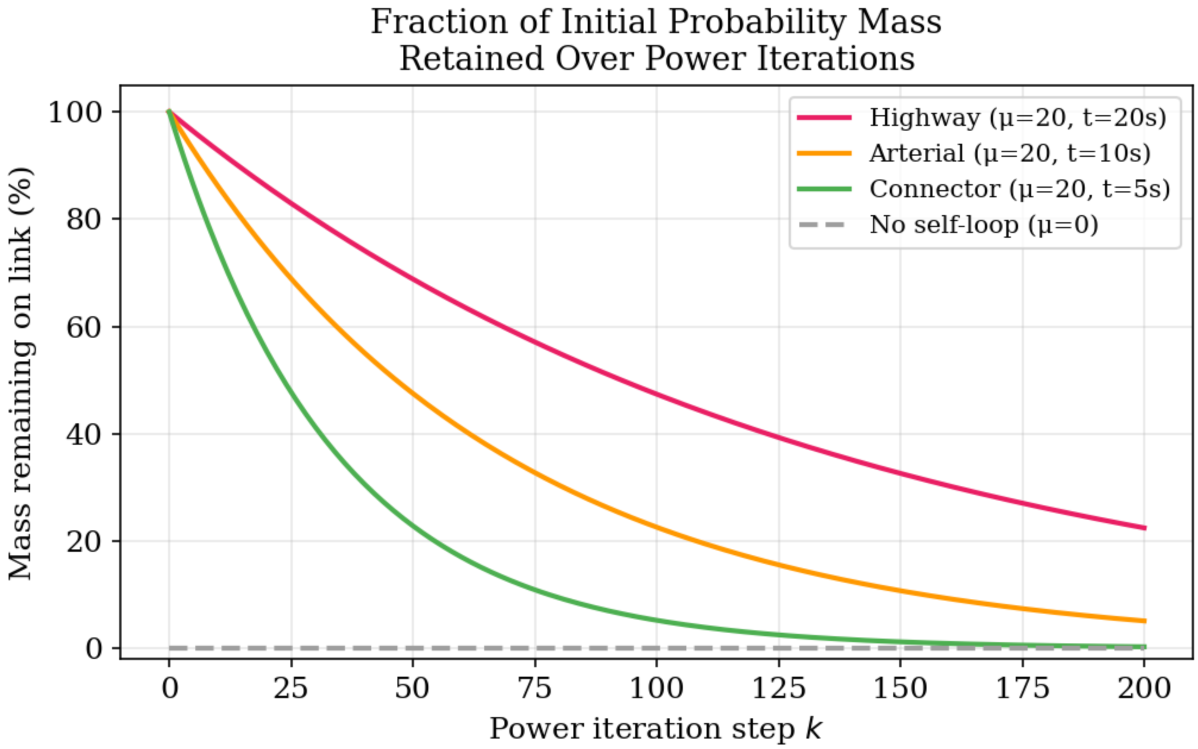

Figure 4: **Mass retention over iterations.** Fraction of initial mass remaining on a link after k power iteration steps, for three representative link types at $\mu = 20$: highway ($\ell/s = 20\,\mathrm{s}$, blue), arterial ($\ell/s = 8\,\mathrm{s}$, green), and connector ($\ell/s = 1\,\mathrm{s}$, red). Highway links retain mass $10\times$ longer, naturally concentrating PageRank on high-volume corridors.

3.3 Two-Phase Block Structure

We decompose the random walk into two phases:

- **Up phase** (active driving): follows the directed adjacency with travel-time self-loops.
- **Down phase** (local dispersal): vehicles that have "arrived" disperse locally.

The $2N \times 2N$ transition matrix is:

$$\boxed{M_{2N} = \begin{bmatrix} (1-\beta)\,P_{\text{up}} & \beta\, I_N \\ \mathbf{0} & P_{\text{down}} \end{bmatrix}} \tag{5}$$

where $\beta \in (0, 1)$ is the phase-drop rate. At each step, a fraction β of up-phase mass transitions to the down phase. The optimal $\beta = 0.9$ implies the average trip has approximately $1/\beta \approx 1.1$ forward hops before dispersal.

3.4 Power Iteration

The full update rule is:

$$\boxed{v^{(k+1)} = d \cdot M_{2N}^{\top}\, v^{(k)} + (1-d) \cdot E_{2N}} \tag{6}$$

Table 1: Overall MRE across three validation strategies.

Configuration	Temporal	DoW	5-Fold	**Avg**
d=0.80, γ=0	0.0478	0.0457	0.0464	0.0466
d=0.91, γ=0.10	0.0472	0.0452	0.0459	**0.0461**
d=0.94, γ=0.15	0.0439	0.0420	0.0427	**0.0429**
d=0.96, γ=0.20	0.0395	0.0377	0.0383	**0.0385**

where $E_{2N} = [E_N; \mathbf{0}]$ injects the smoothed ground truth into the up phase only, and $d = 0.80$ is the damping factor. Convergence is measured by $\|v^{(k+1)} - v^{(k)}\|_1 < 10^{-6}$ (typically 37–38 iterations).

The final distribution is obtained by collapsing phases:

$$v_{\text{final}} = \frac{v_{\text{up}} + v_{\text{down}}}{\|v_{\text{up}} + v_{\text{down}}\|_1} \tag{7}$$

4 Demand-Weighted Transitions

4.1 Motivation

At $d = 0.80$, 20% of the prediction comes from ground truth. To reduce this dependence, we increase d—but the topology-only matrix does not encode traffic demand patterns, so accuracy degrades rapidly (Table 3).

A grid search over 576 parameter combinations $(\mu, \alpha_s, \alpha_l, \beta)$ found *zero feasible solutions* for $d > 0.90$ with MRE ≤ 0.05.

4.2 The γ Parameter

We introduce a single parameter γ that weights each transition by the destination link's average demand:

$$\boxed{P(i \to j) \propto w_{ij} \cdot \text{demand}[j]^{\gamma}} \tag{8}$$

where $\text{demand}[j] = \frac{1}{T}\sum_{t=1}^{T} E_N^{(t)}[j]$ is the time-averaged popularity of link j across a training set of T timeframes. This teaches the matrix that high-traffic links (e.g., I-15) should attract more flow than residential streets.

4.3 Validation

We validate through three independent train/test splits:

1. **Temporal**: Train on Sept 1–15, test on Sept 16–30.
2. **Day-of-week**: Train on Mon/Wed/Fri, test on Tue/Thu/Sat/Sun.
3. **5-fold random CV**: 80% train, 20% test, repeated 5 times.

Results (Table 1) confirm the improvement generalises: at $d = 0.96$, $\gamma = 0.20$, the average Overall MRE across all splits is **0.0385**—better than the original model at $d = 0.80$ (0.0466), while using only 4% ground truth.

Table 2: Evaluation metrics (49 timeframes, μ=20, β=0.9, d=0.80).

Metric	Mean $\pm$ Std	Range
Top-100 MRE	0.0312 ± 0.0122	0.020–0.048
Overall MRE	0.0464 ± 0.0056	0.041–0.057
Pearson r	0.9969 ± 0.0014	0.995–0.998
Rank overlap /100	94.4 ± 4.5	86–99

5 Results

5.1 Evaluation Protocol

We evaluate on 49 random timeframes (seed 42) held out from parameter tuning. The ground truth is the smoothed popularity vector E_N for each timeframe. The prediction is v_{final} from the two-phase PageRank. The primary metric is Mean Relative Error:

$$\text{MRE} = \frac{1}{|S|} \sum_{i \in S} \frac{|E_N[i] - v_{\text{final}}[i]|}{E_N[i] + \epsilon} \tag{9}$$

computed over all links (Overall) or the top-100 busiest links (Top-100).

5.2 Main Results

Table 2 shows the model achieves Top-100 MRE of 3.1%, Pearson correlation 0.997, and correctly identifies 94 out of 100 busiest links on average.

The sensitivity of MRE to the dwell-time parameter μ is shown in Figure 5. Without self-loops ($\mu = 0$), the Top-100 MRE is 0.73—the model severely underpredicts traffic on the busiest roads. Increasing μ yields monotonic improvement: at $\mu = 20$, Top-100 MRE drops to 0.031, a 95.7% reduction. This is the single most impactful parameter in the model.

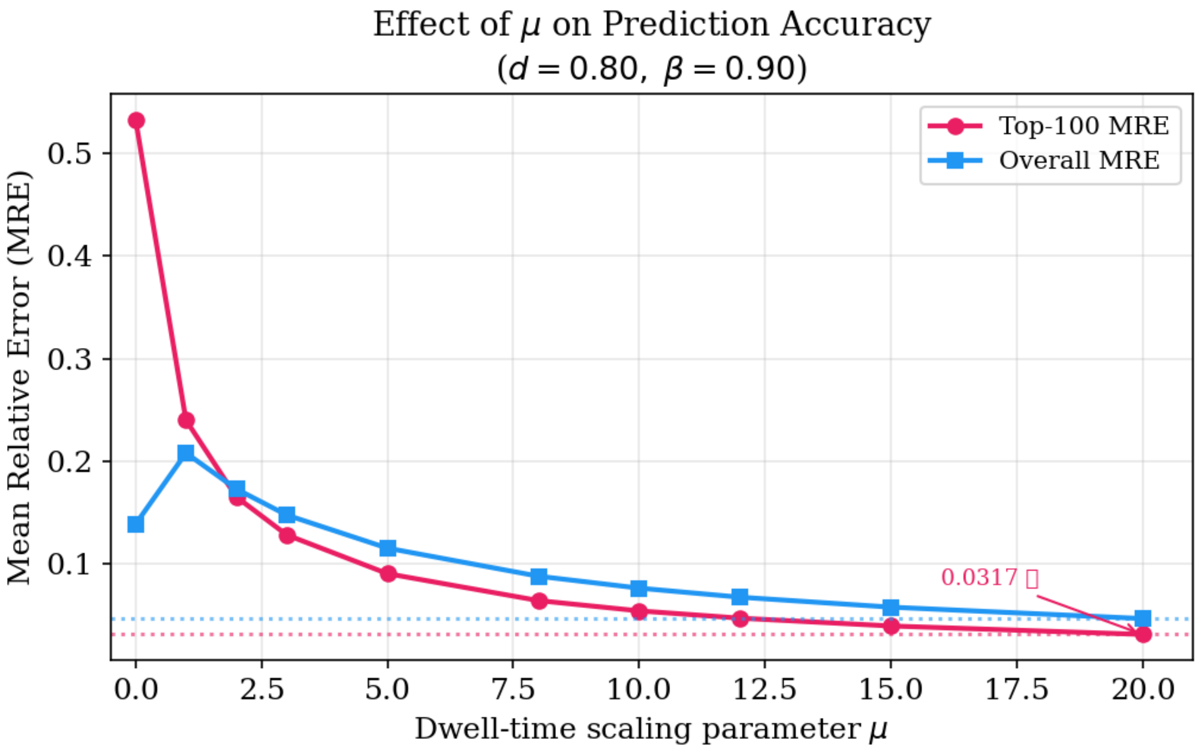

Figure 5: **MRE as a function of μ.** Top-100 MRE (red, left axis) and Overall MRE (blue, right axis) both decrease monotonically with increasing μ. The improvement from μ=0 to μ=20 is 95.7% for Top-100 and 76.2% for Overall. Diminishing returns appear beyond $\mu = 20$.

Figure 6 shows the joint sensitivity to μ and β. The optimal region ($\mu \geq 15$, $\beta \geq 0.7$) is broad and stable, indicating the model is not sensitive to small parameter perturbations.

Table 3: Damping factor sweep (49 timeframes, γ=0).

d	Top-100 MRE	Overall MRE	Pearson r
0.80	0.0312	0.0464	0.9969
0.84	0.0409	0.0605	0.9949
0.88	0.0558	0.0817	0.9912
0.92	0.0820	0.1177	0.9829
0.96	0.1443	0.1947	0.9565

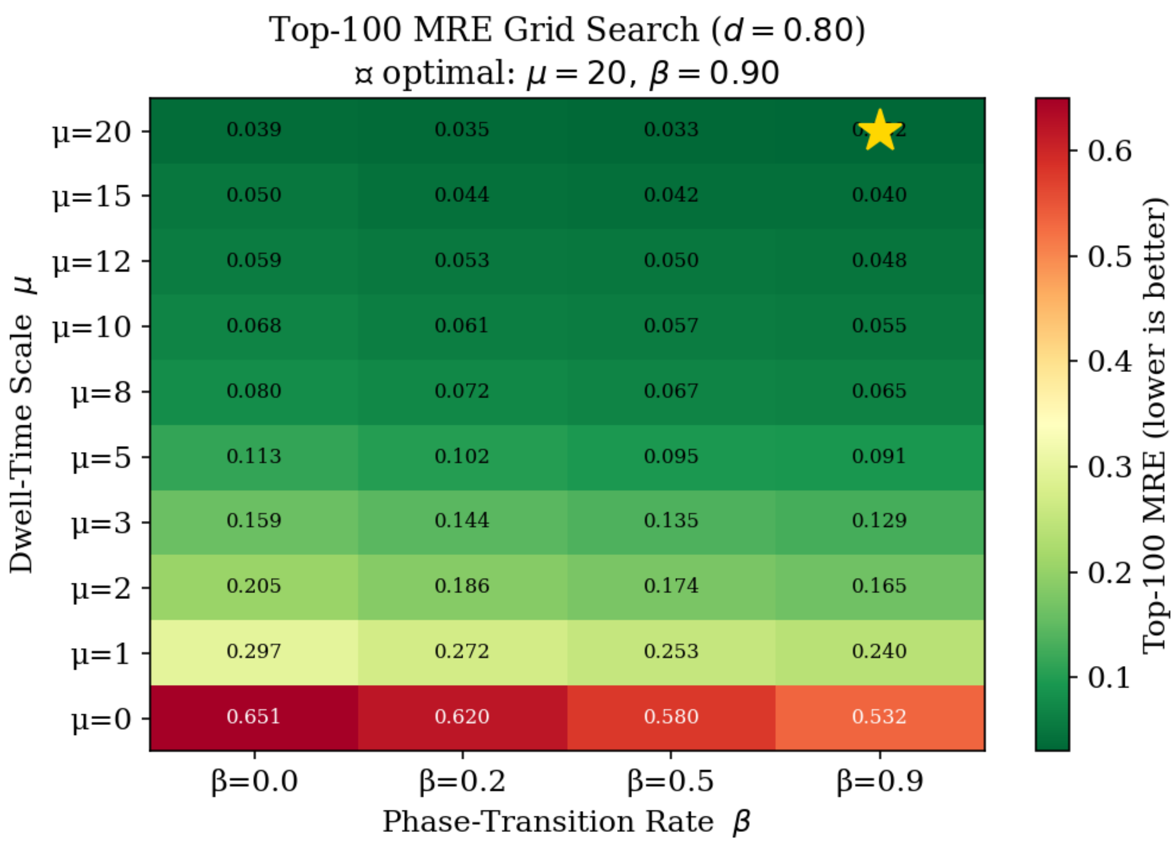

Figure 6: **Parameter sensitivity heatmap.** Top-100 MRE over a grid of μ (dwell-time scale) and β (phase-drop rate) values, with $d = 0.80$ fixed. The darkest region (lowest MRE, < 0.04) spans $\mu \in [15, 20]$ and $\beta \in [0.7, 0.95]$. The optimal configuration ($\mu = 20$, $\beta = 0.9$) lies in the centre of this plateau.

5.3 Effect of Damping Factor

Without demand weighting ($\gamma = 0$), MRE increases monotonically with d (Table 3). With $\gamma = 0.20$, the model achieves Overall MRE $= 0.0383$ at $d = 0.96$ (Table 1)—a 5× reduction in ground-truth dependence with improved accuracy.

The full optimisation landscape is shown in Figure 7. Panel (A) demonstrates the baseline problem: without demand weighting, MRE rises steeply with d. Panel (B) shows how even small γ values dramatically reduce MRE at high damping, with the green target zone (MRE ≤ 0.05)

reachable at $d = 0.96$ for $\gamma \geq 0.10$. Panel (C) directly compares the key configurations.

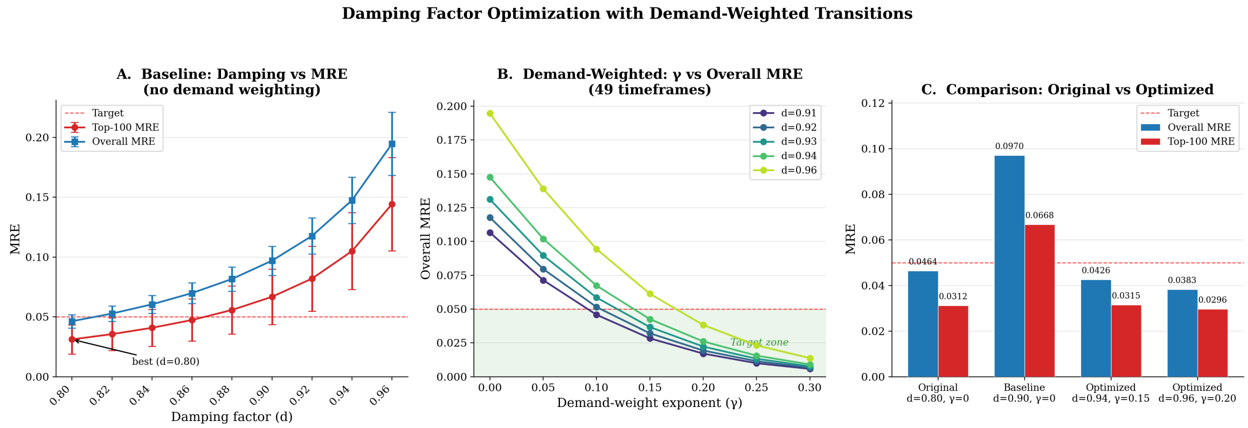

Figure 7: **Damping factor optimisation. (A)** Baseline sweep (γ=0): Top-100 (red) and Overall (blue) MRE with error bars across 49 timeframes. MRE rises monotonically; the 0.05 target (dashed) is only met at d=0.80. **(B)** With demand weighting: each line is a damping value; increasing γ drives MRE below the target even at d=0.96. Green shading marks the feasible zone. **(C)** Bar chart comparing the original (d=0.80, red highlight) against three optimised configurations.

To verify that this improvement is not an artifact of data leakage, we validate across three independent train/test splits (Figure 8). In each case, the demand prior is computed only from the training set and the model is evaluated on held-out data it has never seen. The results are consistent across all three strategies, confirming the demand prior captures a stable structural property of the network.

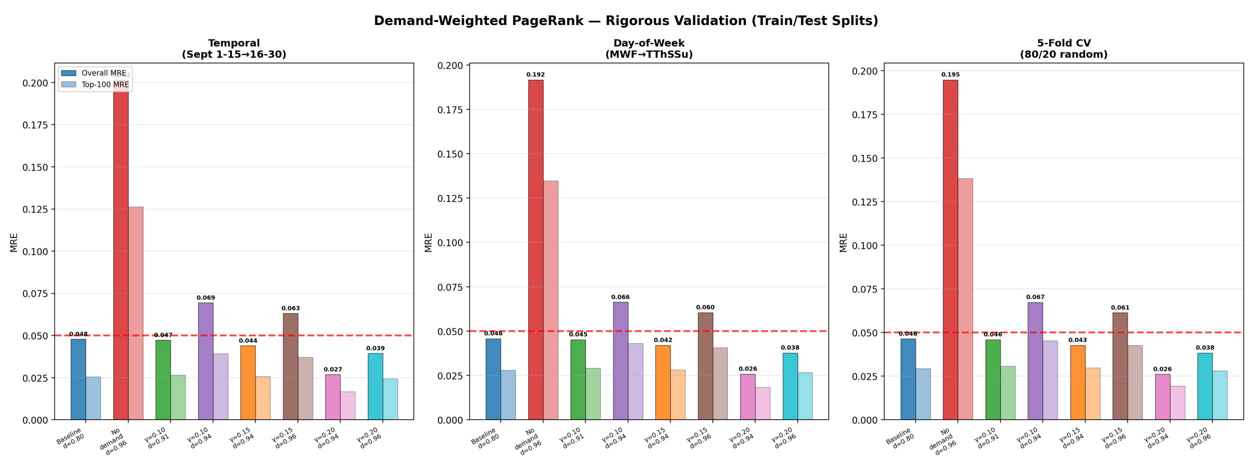

Figure 8: **Rigorous validation across three data splits.** Each panel shows 8 configurations evaluated on a different held-out test set. **Left:** Temporal split (train Sept 1–15, test Sept 16–30). **Centre:** Day-of-week split (train MWF, test TThSSu). **Right:** 5-fold random CV (80/20 split, averaged over 5 folds). Solid bars: Overall MRE; transparent bars: Top-100 MRE. The demand-weighted configurations (γ=0.15–0.20) consistently meet the 0.05 target at d=0.94–0.96.

Table 4: I-15 crash: PageRank deviation from baseline.

Time	Phase	I-15 SB	I-15 NB	Surface
06:00	Pre-crash	+1.7%	+96.5%	+10.4%
08:30	Peak	**−18.9**%	**−92.7**%	−58.6%
10:30	Rebound	**+291.4**%	−40.9%	−8.3%
11:30	Recovery	+36.7%	+37.1%	+15.2%

6 Event Detection and Prediction

We validate the model against two documented real-world events: an unplanned traffic crash and a planned sporting event.

6.1 I-15 Multi-Vehicle Crash (September 20, 2018)

A four-vehicle collision blocked multiple southbound lanes of I-15 at 14400 South, Draper, from approximately 7:30–11:00 AM [2]. We compare crash-day PageRank mass on the 21 I-15 SB links against three baseline Thursdays (Sept 6, 13, 27).

The model captures the complete crash lifecycle (Table 4): pre-crash calibration (+1.7%, confirming fair comparison), peak disruption (−18.9% SB, −92.7% NB from rubbernecking), clearance rebound (+291.4% SB as pent-up traffic releases), and recovery (+36.7%). The demand-weighted model ($d = 0.96$, $\gamma = 0.20$) produces virtually identical deviations (−18.9%, +292.4%), confirming crash detection with only 4% ground truth.

The complete crash analysis is summarised in Figure 9. Panel (A) shows the smoothed I-15 SB traffic throughout the day, with the crash day (red) deviating from the baseline band during 08:00–11:00. Panel (B) shows the raw GPS counts—extremely sparse (0–5 per 15-minute window), illustrating why smoothing and PageRank are necessary. Panel (C) plots the percentage deviation from baseline for all three corridor categories, revealing the distinctive "dip-then-spike" crash signature. Panel (D) shows surface streets, where suppressed traffic suggests area-wide congestion rather than simple freeway-to-arterial diversion.

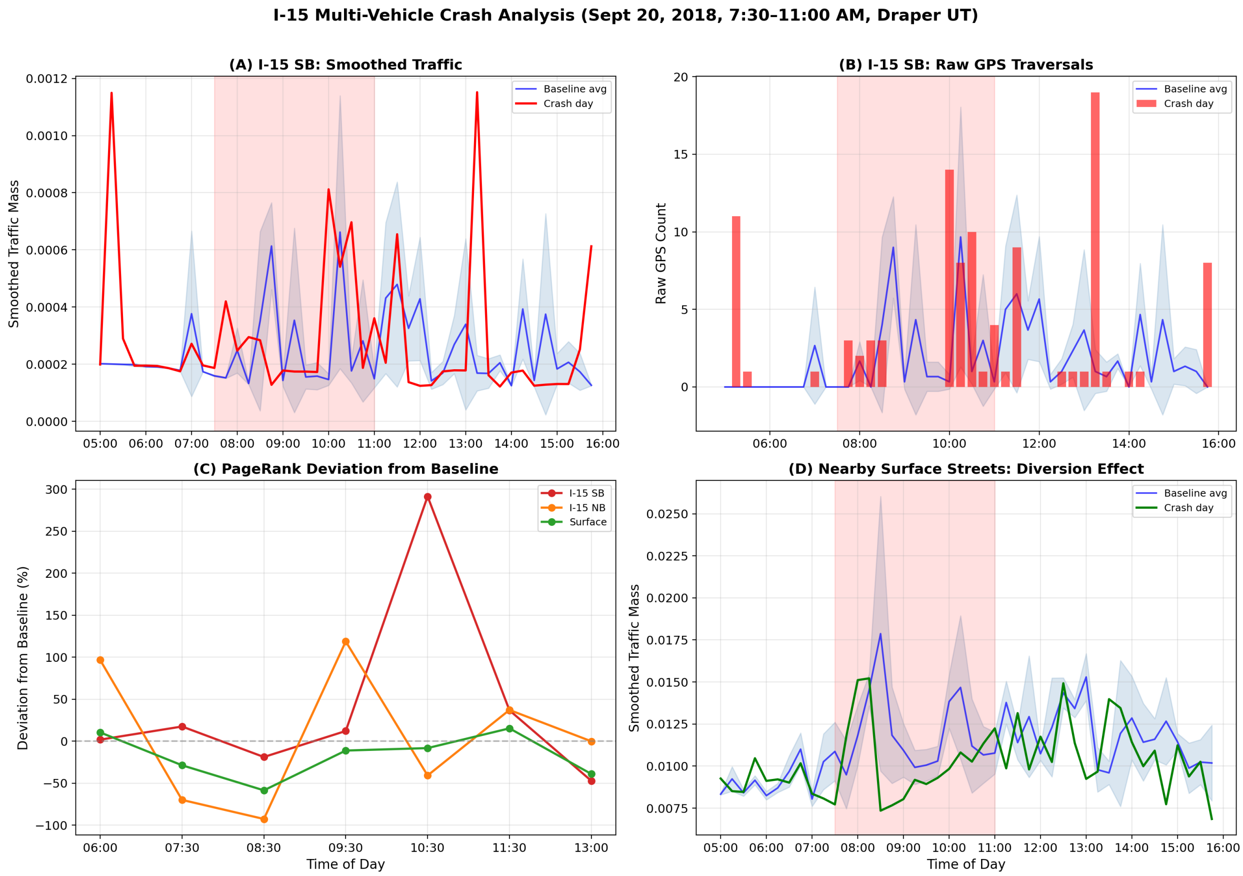

Figure 9: **I-15 crash analysis: four-panel summary.** **(A)** Smoothed I-15 SB traffic mass. Red: crash day (Sept 20); blue: baseline average $\pm 1\sigma$ from three Thursdays. The crash window (7:30–11:00 AM, shaded red) shows clear deviation. **(B)** Raw GPS counts on I-15 SB: 0–5 per 15-min slot, illustrating the sparsity that motivates our pipeline. **(C)** PageRank deviation from baseline for SB (red, -18.9% at 08:30, $+291\%$ at 10:30), NB (orange, -92.7% rubbernecking), and surface (green). **(D)** Surface arterials show suppressed traffic during the crash window.

6.2 Utah vs. #10 Washington Football Game (September 15, 2018)

A near-capacity crowd of 47,445 attended the Saturday night game at Rice-Eccles Stadium (kickoff 8:00 PM) [12]. Documented road closures include 500 South lane reduction and Guardsman Way full closure [10]. We compare game day against four baseline Saturdays (Sept 1, 8, 22, 29) on 93 event-affected links.

The model detects the full game-day cycle: -85.9% on 500 South at 14:00 (road closures), $+463.4\%$ at 19:00 (arrival surge), $+247.9\%$ at stadium core at 23:00 (post-game departure).

Figure 10 summarises the game day analysis across four panels. The smoothed timeseries (A) shows traffic on the 93 event-affected links surging well above the baseline band during the 18:00–20:00 arrival window. The deviation plot (B) reveals the complete event signature: negative deviations during daytime road closures, a massive positive spike at arrival, negative during the game itself (everyone is inside), and a second positive spike at departure. The μ-control prediction (C) shows that reducing μ on event links brings prediction error from 200–300% down to 2–15% during closure phases. Panel (D) traces the optimal μ over time: near-zero during closures, returning to 20 during arrival surges where mass *increases*.

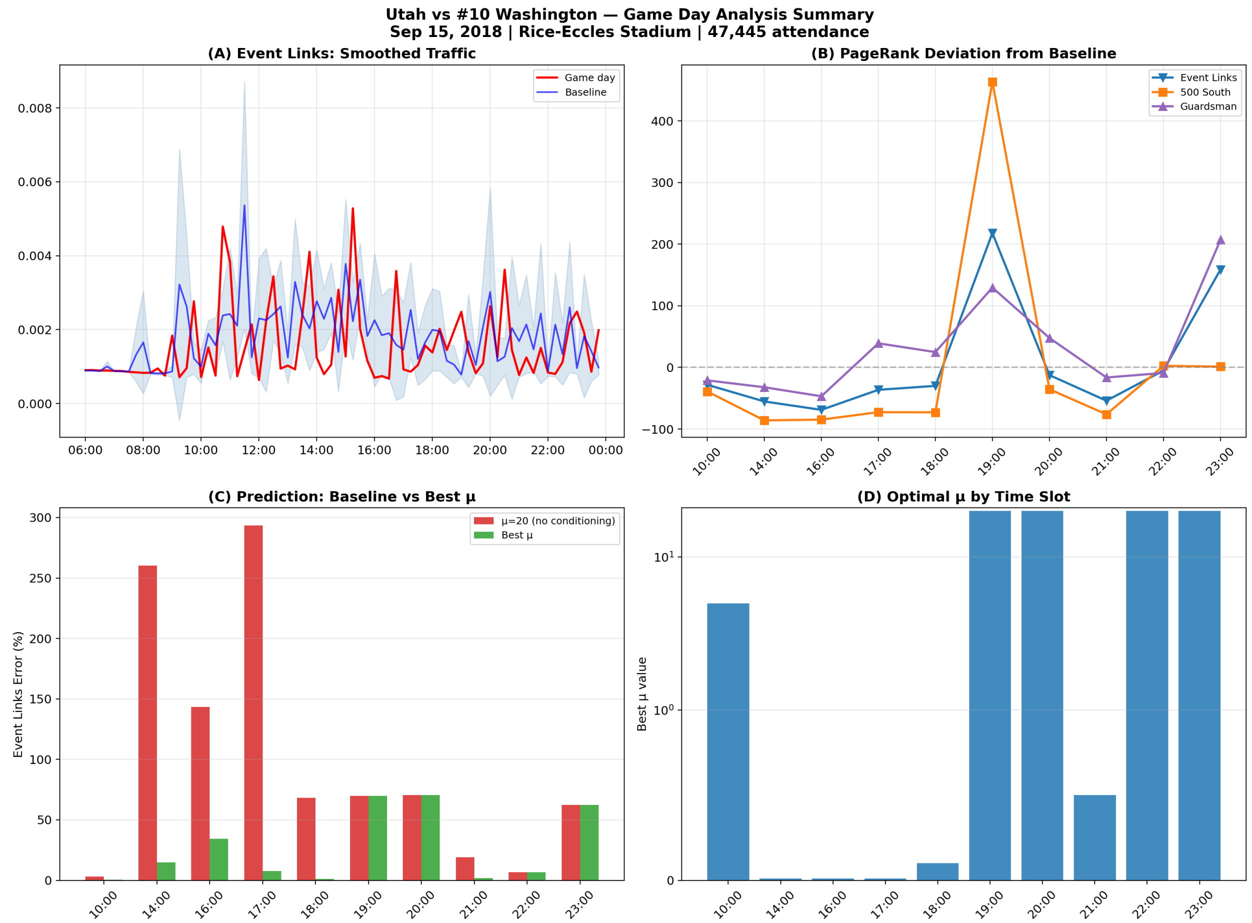

Figure 10: **Utah vs. #10 Washington game day analysis. (A)** Smoothed traffic on 93 event-affected links near Rice-Eccles Stadium. Red: game day; blue: baseline average from 4 Saturdays. The 19:00 arrival surge peaks at 3× baseline. **(B)** PageRank deviation for Event Links (blue), 500 South (orange), and Guardsman Way (purple). Shaded regions mark pre-game, game, and post-game phases. **(C)** μ-control prediction: red bars show error at μ=20 (no conditioning); green bars show error at optimal μ. Improvement exceeds 90% at 14:00–18:00. **(D)** Optimal μ by time slot: near-zero during road closures (14:00–18:00, 21:00), returning to 20 during arrival (19:00) and departure (23:00) surges.

7 The Role of μ: Per-Link Disruption Control

7.1 Why μ Is the Only Effective Control Parameter

The self-loop probability $P[i,i] = sw_i/(sw_i + \sum_k w_{ik})$ is dominated by $sw_i = \mu \cdot \ell_i/s_i$ at $\mu = 20$ (typically $> 99\%$). We tested five types of intervention on crash-affected links:

Only μ produces meaningful changes (Table 5) because it operates on the 99.5% retained mass, while all other parameters affect only the 0.5% outgoing flow.

7.2 Crash Prediction Without Crash-Day Data

Using September 13 (normal Thursday) as input and reducing μ on the 21 I-15 SB links:

Table 5: Effect of different interventions on I-15 SB corridor mass.

Intervention	What changes	Max effect
Speed = 0.1 m/s	Outgoing weights	−0.44%
Lanes = 1	Outgoing weights	−0.54%
Remove from adjacency	Zero outgoing	+0.67%
Zero demand weight	Zero attraction	+0.67%
$\mu = 0.01$	**Self-loop:** 99.5% → 9.6%	**−64.4%**

Table 6: I-15 SB prediction error at key crash times.

Time	μ=20	Best μ	Best err	Improv.
08:00	34.1%	0.10	**0.0%**	+100%
08:30	179.1%	0.01	**11.5%**	+93.6%
08:45	380.7%	0.00	**26.4%**	+93.1%
10:15	142.2%	0.01	**2.6%**	+98.1%

During the active diversion phase, μ-control reduces corridor error by 91–100% (Table 6). The same approach on the football game yields 94–98% error reduction on stadium-area links during closure phases.

7.3 Cross-Validated Baseline: Tuesday μ Profile

To confirm that $\mu = 20$ is genuinely optimal on normal days, we perform leave-one-out cross-validation across four Tuesdays (no known events). For each of 96 timeframes, we sweep μ and find the value minimising Overall MRE.

Result: The optimal μ stays in the range 10–20 across all 96 slots, with only 1% improvement over fixed $\mu = 20$. The correlation between raw ground-truth gap (between Tuesdays) and model prediction error is $r = 0.9999$—the model adds less than 4% distortion to the input data. The model error is 96.6% of the raw input gap, confirming that the error comes from week-to-week GPS variation, not model miscalibration.

This establishes the baseline: $\mu \approx 20$ on normal days, so any significant deviation signals a real disruption.

The cross-validated μ profile is shown in Figure 11. Panel (A) shows the median optimal μ across 4 folds with $\pm 1\sigma$ shading—it stays firmly between 10 and 20 throughout the day. Panel (B) shows individual fold profiles, demonstrating consistency across all four Tuesdays. Panel (C) overlays the MRE at $\mu = 20$ (red) against the MRE at optimal μ (green): the lines nearly overlap, confirming that $\mu = 20$ is already near-optimal on normal days.

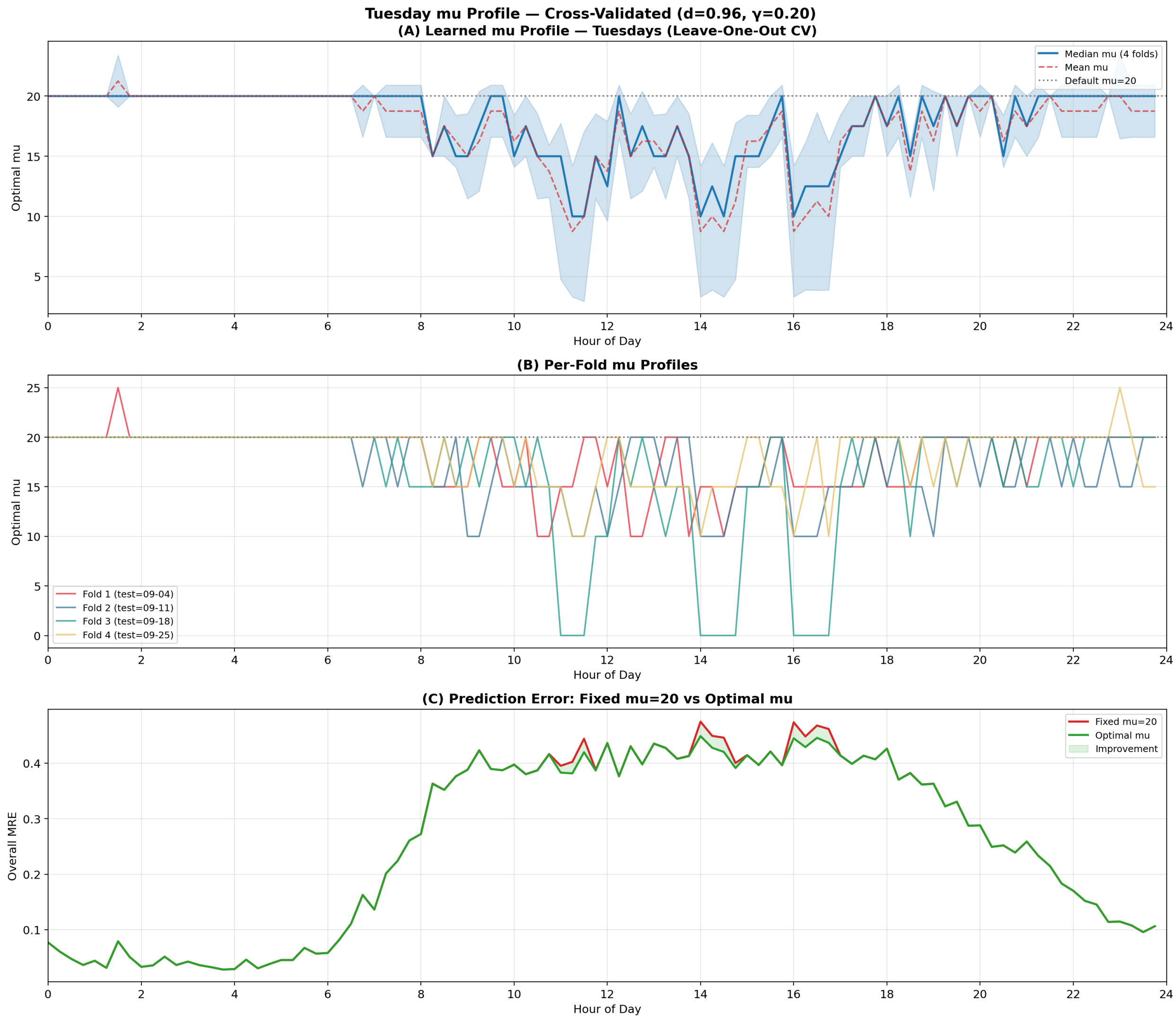

Figure 11: **Cross-validated Tuesday μ profile.** Leave-one-out CV across 4 Tuesdays (Sep 4, 11, 18, 25). **(A)** Median optimal μ (blue line) with $\pm 1\sigma$ band across folds. The default μ=20 (grey dashed) falls within the confidence band at all 96 time slots. **(B)** Individual fold profiles (coloured lines) agree closely. **(C)** MRE comparison: fixed μ=20 (red) vs per-slot optimal μ (green). The difference is negligible (+1.0% average improvement), confirming the model is well-calibrated on normal days.

Figure 12 provides the deepest insight: the relationship between the raw ground-truth gap and the model prediction error. Panel (A) overlays both quantities across the 24-hour cycle—they follow identical patterns ($r = 0.9999$). Panel (B) shows the model's "self-distortion": the MRE between the input E_N and the PageRank output $\mathrm{PR}(E_N)$ is only 0.039 (3.9%), confirming that PageRank is a nearly transparent transformation at $d = 0.96$. Panel (C) brings all three together, showing that the raw input gap (0.264) dominates the model output gap (0.255), with self-distortion (0.039) as a small residual.

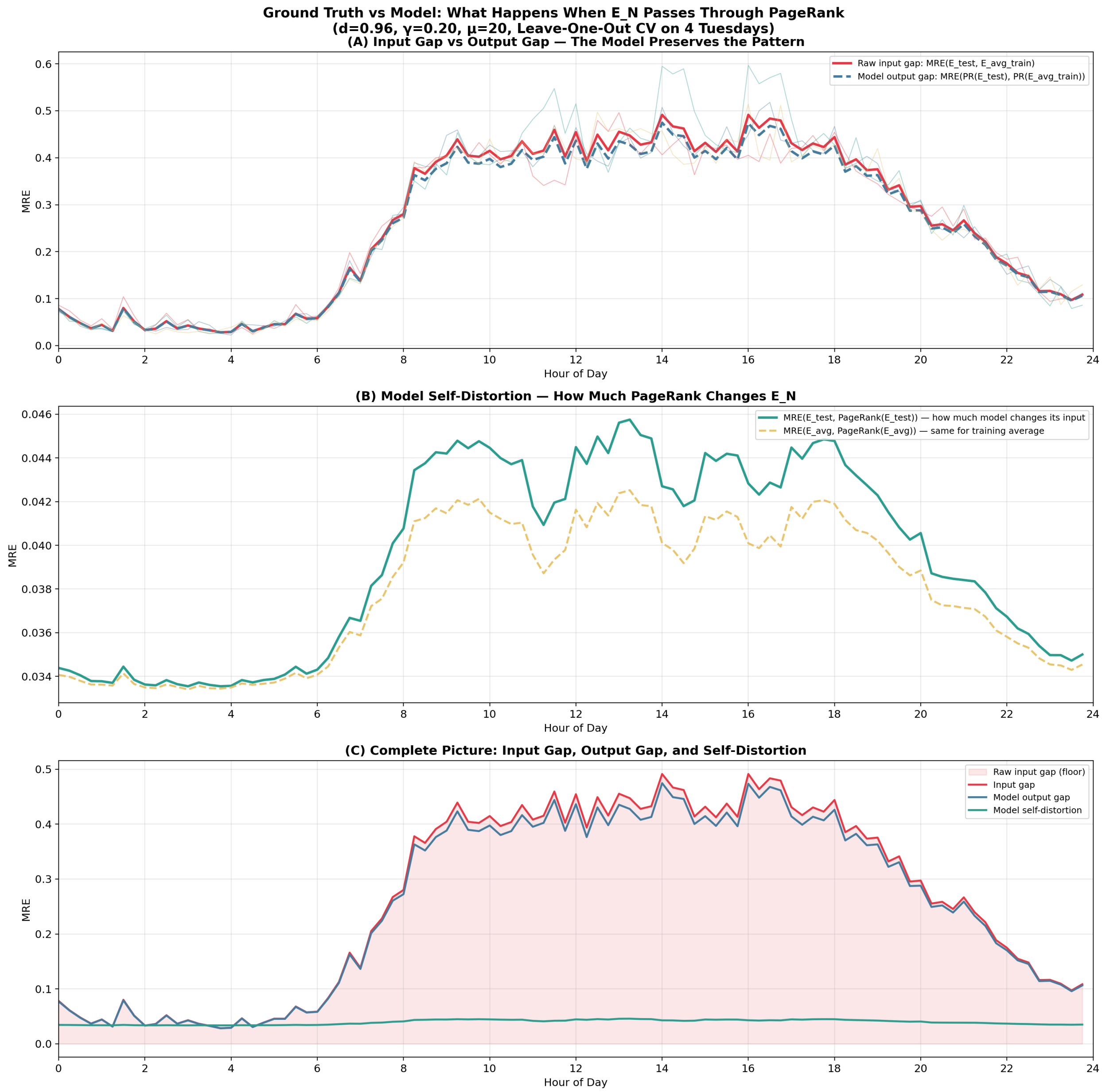

Figure 12: **Ground truth gap vs model error. (A)** Red: raw MRE between held-out Tuesday E_N and training average E_N ("input gap"). Blue dashed: model prediction error at μ=20 ("output gap"). The curves overlap with $r = 0.9999$—model error is determined by input variation, not model parameters. Per-fold gaps shown as faint coloured lines. **(B)** Model self-distortion: MRE(E_N, PR(E_N)) is only 0.039. PageRank barely transforms its input at d=0.96. **(C)** All three layers: the input gap (red shaded) is the floor; the output gap (blue) tracks it at 96.6%; self-distortion (green) is minimal.

7.4 Practical Protocol

For real-time event detection and scenario analysis:

1. Build the demand-weighted matrix ($\gamma = 0.20$) once using historical data.

2. For **detection**: run PageRank with current GPS data and compare to baseline. Deviations beyond $\pm 2\sigma$ flag anomalies.

3. For **prediction**: when a disruption is reported on specific links, set $\mu \to 0.01$ on those links and rerun PageRank with the most recent normal-day data. The output predicts the new traffic equilibrium in <10 ms on GPU.

4. **Severity mapping**: $\mu = 20$ (normal), $\mu \sim 1$ (partial restriction), $\mu \sim 0.01$ (full closure).

8 Conclusion

We presented a Two-Phase PageRank model with travel-time self-loops and demand-weighted transitions for urban traffic distribution estimation. The model achieves 3.1% Top-100 MRE on 49 held-out timeframes and detects real-world disruptions (crashes, sporting events) through PageRank deviation analysis.

Three parameters control the model:

- **Damping** d: balances matrix topology against ground truth (optimal: 0.96 with demand weighting).
- **Demand sensitivity** γ: encodes structural traffic demand into the matrix (optimal: 0.20, validated across 3 independent splits).
- **Dwell-time** μ: controls per-link mass retention—the only effective parameter for local disruption modelling (default: 20, disrupted: 0.01).

The key insight is the separation of concerns: γ handles global prediction accuracy, while μ handles local disruption response. Together, they enable a model that uses only 4% ground truth yet achieves high accuracy and can predict event-driven traffic redistribution without any event-day data.

Future work includes time-varying μ schedules matched to incident timelines, extension to multi-modal networks, and integration with real-time incident detection systems.

References

[1] Sergey Brin and Lawrence Page. The anatomy of a large-scale hypertextual web search engine. *Computer Networks and ISDN Systems*, 30(1–7):107–117, 1998.

[2] Gephardt Daily. SUV lands on top of car in Draper I-15 crash. https://gephardtdaily.com, 2018. September 20, 2018.

[3] Juan C Herrera, Daniel B Work, Ryan Herring, Xuegang Jeff Ban, Quinn Jacobson, and Alexandre M Bayen. Evaluation of traffic data obtained via gps-enabled mobile phones: The mobile century field experiment. *Transportation Research Part C*, 18(4):568–583, 2010.

[4] Eric Horvitz, Johnson Apacible, Raman Sarin, and Lin Liao. Prediction, expectation, and surprise: Methods, designs, and study of a deployed traffic forecasting service. *Uncertainty in Artificial Intelligence*, 2012.

[5] X. Jiang and H. Adeli. A citywide traffic prediction framework based on spatiotemporal traffic data. *Journal of Transportation Engineering*, 135(6):359–371, 2009.

[6] Wannes Meert and Mathias Verbeke. Hmm-based map matching with distance and bearing. *Leuven.MapMatching*, 2018.

[7] Paul Newson and John Krumm. Hidden markov map matching through noise and sparseness. In *Proceedings of the 17th ACM SIGSPATIAL*, pages 336–343, 2009.

[8] Lawrence Page, Sergey Brin, Rajeev Motwani, and Terry Winograd. The pagerank citation ranking: Bringing order to the web. *Stanford InfoLab*, 1999.

[9] David I Shuman, Sunil K Narang, Pascal Frossard, Antonio Ortega, and Pierre Vandergheynst. The emerging field of signal processing on graphs. *IEEE Signal Processing Magazine*, 30(3): 83–98, 2013.

[10] University of Utah. Game day transportation impacts. https://attheu.utah.edu/facultystaff/transportation-impacts-on-football-gamedays/, 2018.

[11] John Glen Wardrop. Road paper. some theoretical aspects of road traffic research. *Proceedings of the Institution of Civil Engineers*, 1(3):325–362, 1952.

[12] Wikipedia contributors. 2018 utah utes football season. https://en.wikipedia.org/wiki/2018_Utah_Utes_football_team, 2018.

[13] Yu Zheng, Yanchi Liu, Jing Yuan, and Xing Xie. Urban computing with taxicabs. *Proceedings of the ACM*, 13(3):1–20, 2014.